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Project Title:	Molecular Genetics of Root Thigmoresponsiveness in Arabidopsis thaliana		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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	Directional cues within the environment (gravity, touch, light etc) guide plant organs' growth by differential regulation of cell expansion. These tropic movements are coupled to endogenous oscillating movements (nutations) whose origin remains poorly understood. Together, these movements allow plant organs to position themselves for optimal performance. To better understand the mechanisms that govern integrated growth responses to multiple stimuli, we have sought and characterized Arabidopsis thaliana mutations that affect root waving and skewing on tilted hard agar surfaces. Such movements result from a combination of gravity, touch, hydrotropic and other surface-derived stimuli, as well as circumnutations. One of these mutations, named wvd2-1, was shown to alter spiral root growth on agar surfaces and anisotropic cell expansion. The WVD2 protein appears to modulate the arrangement of cortical microtubules in expanding cells. This proposal is aimed at understanding how WVD2 and other related proteins modulate anisotropic		

Task Description:	cell expansion and root growth behavior. This will be accomplished by identifying and characterizing proteins that regulate or modulate WVD2 function, and by characterizing seven other WVD2-related proteins in Arabidopsis. Plants constitute an essential component of Advanced Life Support Systems for long-term space-exploration missions. However, during space flight, their growth will be guided by cues other than gravity, including nutations. Therefore, we need to better understand some of the mechanisms that will contribute to guiding organ growth under these conditions, a primary objective of this proposal. Also, there is limited space available for plant growth in spacecrafts. WVD2 may provide a good tool to design reversible engineering strategies that would confer a shorter stature to a broad range of plants, yet would not alter seed production or plant's ability to perceive gravity at their final destination. Hence, a thorough characterization of WVD2-dependent pathways is of direct relevance to the Space Biology Program. The objective of this proposal is to characterize the molecular mechanisms that allow WVD2 and related WDL proteins to control root growth behavior in response to mechanical stimuli, anisotropic cell expansion and plant morphogenesis. Our long-term goal is to use the information that will derive from this project to engineer shorter and stockier plants than wild type, which can still produce normal levels of seeds, but are better suited for growth in the limited confines of bioregenerative life support systems during long-term space exploration missions, or that better resist to lodging by wind and rain on Earth or other planets. The WVD2 and WDL genes appear to function not only as key regulators of root growth behavior in response to mechanical stimuli, but also as major determinants in the control of anisotropic cell expansion in all organs of the plant. Therefore, a better understanding of these genes is likely to provide important new insights into how mechanical information in the environment controls cell expansion, and how this affects overall plant morphogenesis, a process of fundamental importance for agricultural production and plant adaptation to its environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Root-growth behavior conditions the ability of plants to populate and grow in very heterogenous and diverse environments. Hence, a better understanding of the molecular mechanisms that govern this process is likely to provide the tools needed to engineer plants that are better able to colonize extreme environments on Earth, and on other planets. At the same time, controlled anisotropic cell expansion is an essential component of plant morphogenesis and behavior in response to directional environmental stimuli. Our studies have allowed the identification of an important new dominant regulator of this process. Because over-expression of WVD2 results in shorter plants with thicker organs, it may be possible to use this engineering strategy to generate crop plants that are more resistant to lodging by wind and rain, as well as plants that are better suited for utilization in spacecrafts, where limited amounts of space are available for plant growth.
	1. Introduction: As discussed in previous annual reports, wvd2-1 is a dominant mutation that results from over-expression of a wild type 23 kD protein containing a novel conserved KLEEK domain with similarity to the TPX2 pfam motif found in several plant and animal unconventional microtubule binding proteins. In Arabidopsis, 7 other genes are predicted to encode KLEEK-containing proteins (WDL1 – 7). wvd2-1 seedlings display an inability to develop root waves and opposite root skewing relative to wild type on tilted hard agar surfaces, enhanced spiral organs growth, altered anisotropic cell expansion and enhanced sensitivity to osmotic shock, mechanostimulation and light. These phenotypes are accompanied by altered cortical microtubule alignment in epidermal cells of the elongation zone and in peripheral cap cells (Yuen et al., 2003), and also occur in plants over-expressing any one of the 7 WDL genes. Together, these data suggest that WVD2 and other WDL genes have at least partially overlapping functions. We also reported last year that E. coli - expressed GST-tagged WVD2 protein binds to in vitro polymerized microtubules and promotes their bundling, suggesting that WVD2 regulates anisotropic cell expansion and root growth behavior by interacting with, and promoting the bundling of, cortical microtubules in expanding cells. This year, we further refined these in vitro microtubule binding and bundling experiments to ascertain the microtubule-binding and -bundling activity of WVD2, and expanded our functional, expression and localization studies of other WDL genes. As summarized below, current preliminary data suggest that functional redundancy is only partial in this family of fascinating plant-specific proteins, with accumulating evidence for functional and/or expression specialization for individual members. 2. WVD2 promotes the bundling of microtubules in vitro, and modifies the dynamic instability of cortical microtubules in vivo. To better elucidate the role of WVD2 in the regulation of microtubules organization and/or dynamic instability, we first verified that the GST moiety attached to the purified WVD2 protein used in our in vitro studies was not responsible for some of the observed effects (microtubule binding and/or bundling activities). We generated a new epitope-tagged version of WVD2 by fusing the corresponding cDNA to the maltose binding protein sequence of pMAL-c2x (New England Biolabs), generating an MBP-WVD2 construct that was expressed in E. coli. The corresponding protein was purified and added to in vitro assembled, rhodamine-labeled microtubules, using protocols described by the supplier (Cytoskeleton, Inc, Denver, CO). Results demonstrated that MBP-WVD2 is as efficient at promoting microtubule bundling in vitro as is the GST-WVD2 protein used in our previous experiments. This important result strongly supports a role for WVD2 in binding to microtubules and promoting their bundling. To further investigate in vivo function of WVD2, we quantified the orientation of cortical microtubules in expanding cells of the lateral root cap of wild-type Ler, WVD2-over-expressing wvd2-1, and knockout wvd2-2 mutant seedlings. The results indicate that WVD2 over-expression randomizes the orientation of cortical microtubules in these cells. Furthermore, the cortical microtubules in wvd2-1 lateral cap cells appeared thicker than those of wild type or wvd2-2 mutant roots, supporting a role for WVD2 in microtubule bundling. To analyze the effect of WVD2 on the dynamic properties of cortical microtubules in live cells, we determined the outcome of microtubule interactions in eGFP:TUB1-7-expressing wild-type and wvd2-1 mutant seedlings, using time-lapse fluorescent microscopy. In plant cells, microtubules progressing in a collision course can either continue growing straight after contact (crossover), depolymerize at the previously growing end (catastrophe), or change direction and grow close to (parallel) each other (zippering) (Dixit and Cyr, 2004). Results indicate that WVD2 over-expression promotes increased zippering in detriment of crossover or catastrophe outcomes. If WVD2 affects the dynamic properties of cortical microtubules in expanding cells, we expect an increased number of deformed trichomes in wvd2-1 compared to wild type, as discussed in (Abe and Hashimoto, 2005). Trichomes with 1, 2, 3 or 4 branches were counted in rosette leaves of 21-day-old wild type and wvd2-1 mutant plants grown on hard-agar media. wvd2-1 leaves displayed a larger proportion of trichomes with fewer branches (1 or 2) than wild-type leaves (3). Taken together, the data accumulated last year are compatible with WVD2 functioning in vivo by modulating the

Task Progress:	dynamic instability of cortical microtubules. 3. Could WVD2 and/or some WDL proteins modulate aspects of brassinosteroid signal transduction in plants? As discussed in last year's progress report, wvd2-1 seedlings display enhanced sensitivity to high light treatments. When light intensity is raised above 120µE.m-2.s-1, mutant seedlings become pale, and their epidermal cells bulge out. This and other observations suggest a possible link between the brassinosteroid-response pathway and WDL-mediated regulation of the dynamic properties of cortical microtubules in expanding cells. Work is in progress to investigate this possible link. 4. Expression and protein localization studies of WDL1-WDL7. Last year, we reported having generated transgenic plants containing constructs that fuse the WDL genes to GFP, in frame, under their native promoters, or under the strong CaMV 35S promoter (for over-expression), along with constructs fusing the native WDL promoters to GUS. These constructs were aimed at testing expression patterns, protein localization and the phenotypic effects of WDL over-expression in Arabidopsis. We also reported that over-expression of WDL-GFP fusions resulted in phenotypes that are similar to those derived from over-expression of the wild-type WDL proteins, suggesting the GFP moiety in these constructs does not affect protein function. This year, we propagated multiple independent transformants for each construct, and generated stable transgenic lines that are currently being investigated for GUS or GFP expression patterns, and for GFP localization within expressing cells. Preliminary results using WDL1p-GUS and WDL4p:GUS expression constructs indicate complementary expression patterns for these two genes. We also reported having isolated knockout mutants in all WDL genes, and observing initial phenotypes for seedlings grown on media containing drugs that destabilize the microtubules, such as propyzamide or aryzalin. We have continued the phenotypic analysis of these mutants, and have initiated experiments aimed at generating double mutants within this gene family. Interestingly, preliminary results suggest that the wdl1-4 wdl6-1 double mutant might be lethal. Furthermore, segregation analysis from crosses between wdl1-4 and wdl4-1, or between wdl1-4 and wdl6-1, indicate semi-sterility associated with the corresponding double heterozygous plants. Experiments are in progress to determine the developmental basis for semi-sterility in these backgrounds. 5. Summary. Last year, we accumulated in vitro and in vivo evidence supporting a role for WVD2 in controlling the dynamic properties of cortical microtubules in expanding cells of Arabidopsis. We also generated material aimed at studying the biological role of other WDL genes and the corresponding proteins in plant growth and development, including optimizing protocols and strains aimed at testing possible interactions with the brassinosteroid-response pathway, propagating transgenic lines expressing reporter constructs aimed at defining expression patterns and protein localization for all WDL genes present in Arabidopsis, and generating single and double mutants for further phenotypic characterization. Research during the last few months covered by this grant will be aimed at characterizing this material. These analyses should shed some light on the partially overlapping functions played by these genes in plant growth, development and response to their mechanical and biotic environment. In the long run, the corresponding information will help designing and engineering shorter and stockier plants for utilization in the limited confines of Bioregenerative Life Support systems during long-term space exploration missions. 6. Cited literature Abe, T., and Hashimoto, T. (2005). Altered microtubule dynamics by expression of modified alpha-tubulin protein causes right-handed helical growth in transgenic Arabidopsis plants. Plant J 43, 191-204. Dixit, R., and Cyr, R. (2004). Encounters between dynamic cortical microtubules promote ordering of the cortical array through angle-dependent modifications of microtubule behavior. Plant Cell 16, 3274-3284. Hirabayashi, S., Matsushita, Y., Sato, M., Oh-I, R., Kasahara, M., Abe, H., and Nyunoya, H. (2004). Two proton pump interactors identified from a direct phosphorylation screening of a rice cDNA library by using a recombinant BRI1 receptor kinase. Plant Biotech 21, 35-45. Perrin, R.M., Wang, Y., Yuen, C.L., Will, J., and Masson, P.H. (2006). WVD2 and WVD2-Like 1 are novel microtubule-associated proteins in Arabidopsis thaliana. To be submitted within a couple of weeks or so. Yuen, C., Pearlman, R., Silo-Suh, L., Hilson, P., Carroll, K., and Masson, P. (2003). WVD2 and WDL1 modulate helical organ growth and anisotropic cell expansion in Arabidopsis. Plant Physiol 131, 493-506.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Perrin RM, Will J, Wang Y, Yuen CY, Masson PH. "WVD2, a novel microtubule binding protein affecting plant anisotropic cell expansion, morphology and helical handedness." 16th International Conference on Arabidopsis Research, Madison, WI, June 15, 19, 2005. Book of Abstracts, 16th International Conference on Arabidopsis Research, University of Wisconsin-Madison, June 2005. , Jun-2005
Abstracts for Journals and Proceedings	Wang Y, Will J, Perrin RM, Yuen CY, Masson PH. "WVD2-Like Proteins Regulate Plant Growth Behavior and Anisotropic Cell Expansion." 16th International Conference on Arabidopsis Research, Madison, WI, June 15-19, 2005. Book of Abstracts, 16th International Conference on Arabidopsis Research, University of Wisconsin-Madison, June 2005. , Jun-2005
Articles in Peer-reviewed Journals	Yuen CYL, Sedbrook JL, Carroll KL, Masson PH. "Loss-of-Function Mutations of ROOT HAIR DEFECTIVE3 Suppress Root Waving, Skewing and Epidermal Cell File Rotation in Arabidopsis thaliana (L.) Heinh. " Plant Physiol. 2005 Jun; 138(2): 701-14. PMID: 15908600 , Jun-2005
Articles in Peer-reviewed Journals	Perrin RM, Young LS, Murthy UMN, Harrison BR, Wang Y, Will JL, Masson PH. "Gravity signal transduction in primary roots. " Ann Bot (Lond). 2005 Oct;96(5):737-43. PMID: 16033778 , Oct-2005